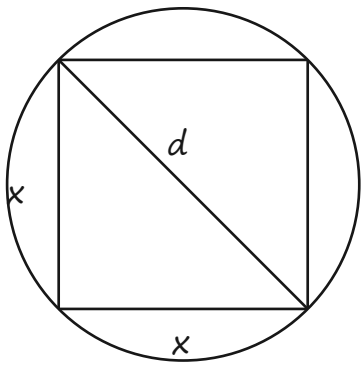
8 A square, with sides of length x cm, is inside a circle.
Each vertex of the square is on the circumference of the circle.

The area of the circle is 49 cm^2.

Work out the value of x.
Give your answer correct to 3 significant figures.



Circle area: $\pi r^2 = 49$

$$r^2 = \frac{49}{\pi}$$

Diameter $d = 2r$

$$d^2 = 4r^2 = \frac{196}{\pi}$$

1 mark
Use the circle area to find r or d

The square diagonal is a diameter.

$$x^2 + x^2 = d^2$$

$$2x^2 = \frac{196}{\pi} \qquad \text{so } x^2 = \frac{98}{\pi}$$

1 mark
Use Pythagoras

1 mark
Rearrange for x^2

$x = \sqrt{(98 \div \pi)} = 5.58519...$ cm

1 mark
5.59 cm to 3 significant figures

5.59

(Total for Question 8 is 4 marks)